

σ -algebra ^(σ -field) :

$$X, \quad \Sigma \subseteq P(X)$$

$$1. X \in \Sigma.$$

$$2. \Sigma \text{ closed under complementation. } (A \in \Sigma \Rightarrow X \setminus A \in \Sigma)$$

$$3. \text{ closed under countable } \# \text{ of unions. } \left(\underbrace{A_1, \dots}_{\text{countable}} \in \Sigma \Rightarrow \bigcup A_i \in \Sigma \right)$$

$$X = \{1, \dots, 6\}$$

$$\Sigma = \{ \{1, 3, 5\}, \{2, 4, 6\}, \{1, 2, \dots, 6\}, \emptyset \}$$

Measurable Space.

(X, Σ)

(Ω, Σ, P)

set of outcomes (samples)

Probability Space.

measure defined on Σ .

$$P(\Omega) = 1$$

$$P(\emptyset) = 0$$

set of events $P(\cup A_i)$

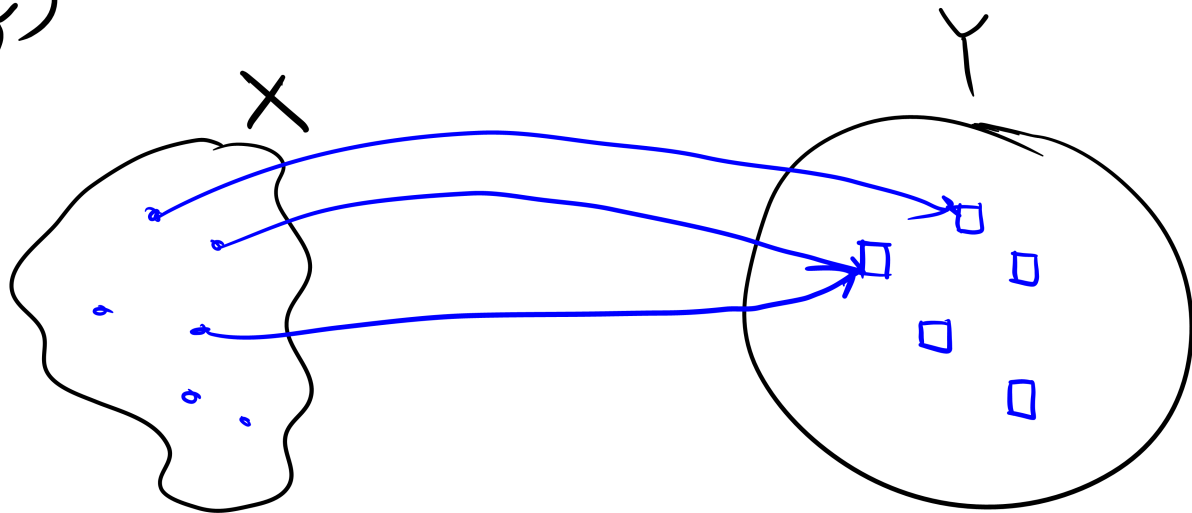
$$= \sum_i P(A_i)$$

(A_i 's are disjoint)

Random Variable.

Measurable functions.

(X, Σ_X) (Y, Σ_Y)



$f: X \rightarrow Y$

$$\forall B \in \Sigma_Y; f^{-1}(B) = \{ \omega \in X \mid f(\omega) \in B \} \in \Sigma_X$$

$$X : (\Omega, \mathcal{F}) \longrightarrow (\mathbb{R}, \mathcal{B})$$

eg. $\uparrow$ $\uparrow$ $\searrow$ $\sigma(\text{all open intervals})$
 probability of $X \geq 5$.

$$P(X \in B)$$

CDF $P(X \leq t) = F_X(t)$

Tower property

1. if $\mathcal{G} \subseteq \mathcal{H} \subseteq \mathcal{F}$

$$\mathbb{E}[\mathbb{E}[X|\mathcal{H}]|\mathcal{G}] = \mathbb{E}[X|\mathcal{G}]$$

1.2: Law of total expectation.

$$\mathbb{E}_Y[\mathbb{E}_X[X|Y]] = \mathbb{E}_X[X]$$

2. Jensen Ineq.

f convex $f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$

3. Integrated tail : $\mathbb{E}(X) = \int_0^{\infty} P(X > t) dt$

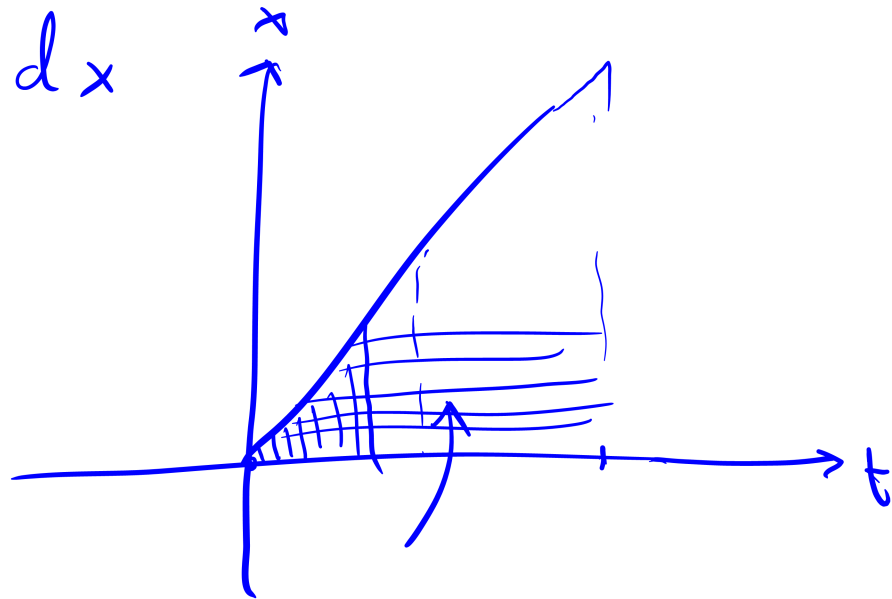
$$E[X] = \int_{-\infty}^{\infty} x f(x) dx =$$

$$P(X \geq t) = \int_t^{\infty} f(x) dx \quad \int_{-\infty}^{\infty} \overbrace{\int_t^{\infty} f(x) dx}^{\text{arrow}} dt$$

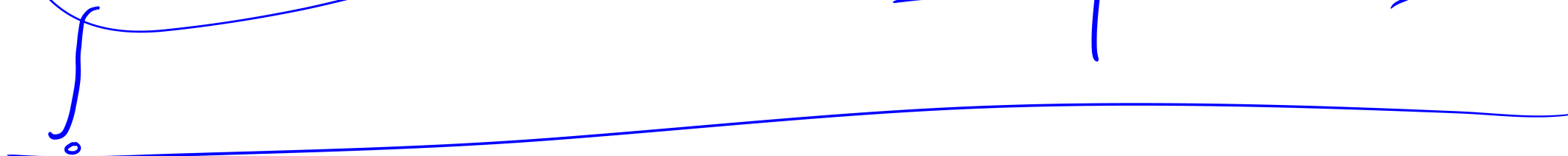
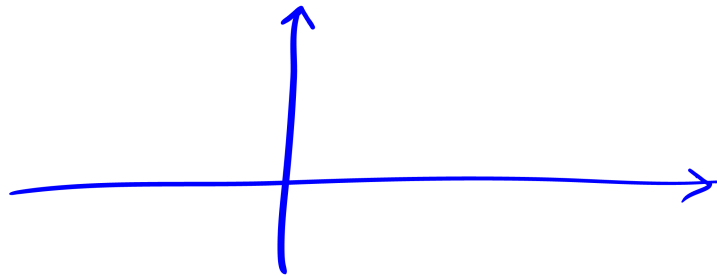
$$x = \int_0^x dt$$

$$\int_{-\infty}^{\infty} \int_0^x f(x) dt dx$$

$$\int_0^x dt = \int_{\int_0^x}^{\infty} \int_0^x$$



$$\int_0^x f(t) dt = F(x)$$



MGFs.

$$M_X(s) : \mathbb{R} \rightarrow [0, \infty)$$

$$M_X(s) = E[e^{sX}]$$

$$X \sim \text{Exp}(\lambda) \Rightarrow M_X(s) = \begin{cases} \frac{\lambda}{\lambda - s} & s < \lambda \\ \infty & \text{o.w.} \end{cases}$$

$$X \sim N(0, 1)$$

$$M_X(s) = \int_{-\infty}^{\infty} e^{sx} f(x) dx = \int_{-\infty}^{\infty} e^{sx} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$$

$$\frac{\exp(s^2/2)}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2}(x-s)^2\right) dx = \exp(s^2/2)$$

$$\rightarrow I = \int_{-\infty}^{\infty} \exp\left(-\frac{x^2}{2}\right) dx = \sqrt{2\pi}$$

$$\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)$$

PDF $N(0,1)$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp\left(-\frac{x^2+y^2}{2}\right) dx dy = I^2$$

Properties:

$$\begin{aligned} 1. \text{ if } Y = aX + b : \quad M_Y(s) &= \mathbb{E}[e^{saX}] \cdot e^{sb} \\ &= e^{sb} M_X(sa) \end{aligned}$$

$$2. \text{ if } X \perp\!\!\!\perp Y : \quad M_{X+Y}(s) = M_X(s) M_Y(s)$$

$$X \sim N(\mu, \sigma^2) \Rightarrow M_X(s) = e^{\frac{s^2 \sigma^2}{2} + s\mu}$$

$$b = \mu$$

$$a = \sigma$$

Applications:

$$\frac{d^m M_X(s)}{ds^m} \xrightarrow{\text{at } s=0} E[X^m]$$

Thm. Inversion Theorem

if $M_X(s)$ is finite $[-a, a]$. $\Rightarrow M_X(s)$ defines a CDF for X .

then if $M_Y(s) = M_X(s) < \infty \Rightarrow X \stackrel{d}{=} Y$.

Example. $X \sim N(\mu_x, \sigma_x^2)$
 $X \perp\!\!\!\perp Y$ $Y \sim N(\mu_y, \sigma_y^2) \Rightarrow X + Y$

$$M_{Y+X}(s) \underset{X \perp\!\!\!\perp Y}{=} M_X(s) M_Y(s) = \exp\left(\frac{s^2}{2} \underbrace{(\sigma_x^2 + \sigma_y^2)}_{\sigma^2} + s \underbrace{(\mu_x + \mu_y)}_{\mu}\right)$$

$$= \exp\left(\frac{s^2 \sigma^2}{2} + s \mu\right)$$

$$\Rightarrow X + Y \sim N(\mu_x + \mu_y, \sigma_x^2 + \sigma_y^2)$$

Multi-variate version

$$M_{X_1, \dots, X_n}(s_1, \dots, s_n) = E[e^{s_1 X_1 + \dots + s_n X_n}]$$

Types of Convergence.

1. (strongest) Almost Sure Conv. $X_n \xrightarrow{\text{a.s.}} X$

$$P(\{\omega \in \Omega \mid \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\}) = 1$$

2. Convergence in probability

$$X_n \xrightarrow{P} X \quad \text{if} \quad \forall \varepsilon > 0 : \lim_{n \rightarrow \infty} P(|X_n - X| > \varepsilon) = 0$$

$\exists n$ sufficiently large.

3. (weakest) In distribution Convergence.

$$X_n \xrightarrow{d} X \quad \text{if} \quad F_{X_n}(x) \rightarrow F_X(x) \quad \text{for every continuity of } F_X.$$

a.s. $\Rightarrow$ In Prob. $\Rightarrow$ In dist.

e.g. SLLN $\overline{X}_n \xrightarrow{\text{a.s.}} \mathbb{E}[X]$

e.g. $\overbrace{01001} \quad \overbrace{0010} \quad \dots$
 $\nearrow \quad \searrow$
 $x_0 \quad x_1$

$$\lim_{n \rightarrow \infty} P(|X_n - 0| > \varepsilon) = 0 \quad \text{In prob.}$$

$$P\left(\lim_{n \rightarrow \infty} X_n = X\right) \quad \text{a/s.}$$

$\swarrow \quad \nearrow$
 lim does not exists.

eg.

$$\{A_n\}_{n=1}^{\infty} \quad P(A_n) = \frac{1}{n} \rightarrow \frac{1}{n^2}$$

$$X_n = \mathbb{1}[A_n]$$

$$\lim_{n \rightarrow \infty} P(|X_n - \overset{0}{X}| > \varepsilon) = 0$$

$\underbrace{\hspace{10em}}_{\frac{1}{n}}$

a.s.

$$\mathbb{P} \left(\lim_{n \rightarrow \infty} \mathbb{1}[A_n] = 0 \right) \leftarrow \text{not a.s.}$$

$$\sum \mathbb{P}(A_n) = \sum \frac{1}{n} = \infty$$

e.g. $U \sim \text{unif}[0,1]$

$$X_{2k} = \mathbb{1} \left[U \leq \frac{1}{2} \right]$$

$$X_{2k+1} = \mathbb{1} \left[U > \frac{1}{2} \right]$$

$$|X_{2k} - X_{2k+1}| = 1$$

$$X_n \xrightarrow{d} \text{Bernoulli} \left(\frac{1}{2} \right)$$

Continuous Convergence Theorem

1. If $X_n \xrightarrow{\text{a.s.}} X$, $g: \mathbb{R} \rightarrow \mathbb{R}$ continuous $\Rightarrow g(X_n) \xrightarrow{\text{a.s.}} g(X)$

2. If $X_n \xrightarrow{P} X$, $g: \mathbb{R} \rightarrow \mathbb{R}$ continuous $\Rightarrow g(X_n) \xrightarrow{P} g(X)$

Lemma 1. (Borel Cantelli) ✓

$\{A_n\}_{n=1}^{\infty}$ If $\sum P(A_n) < \infty$

$$\Rightarrow P\left(\limsup_{n \rightarrow \infty} A_n\right) = 0$$

$$\limsup_{n \rightarrow \infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{N=n}^{\infty} A_N$$

Lemma 2. $X_n \xrightarrow{\mathbb{P}} X$, $\exists$ subsequence $\{X_{n_k}\}$
such that $X_{n_k} \xrightarrow{\text{a.s.}} X$

Proof:

$$A_k := \{ |X_{n_k} - X| > 2^{-k} \}$$

$$\lim_{k \rightarrow \infty} \mathbb{P}(|X_{n_k} - X| > 2^{-k}) = 0$$

$$\mathbb{P}(A_k) = \mathbb{P}(|X_{n_k} - X| > 2^{-k}) \leq 2^{-2k}$$

$$\Rightarrow \sum \mathbb{P}(A_k) < \infty$$

$$\mathbb{P}(A_k \text{ i.o.}) = 0$$

$$\forall \omega \in \Omega ; \quad \exists K(\omega) \quad \text{s.t.} \quad \forall k \geq K(\omega)$$

$$|X_{n_k}(\omega) - X(\omega)| \leq 2^{-k}$$

$$k \rightarrow \infty \implies X_{n_k}(\omega) \rightarrow X(\omega) \implies X_{n_k} \xrightarrow{\text{a.s.}} X$$

Proof of thm.:

$\{X_{n_k}\}$ subsequence of $\{X_n\}$

$$X_{n_k} \xrightarrow{P} X \implies X_{n'_k} \xrightarrow{\text{a.s.}} X \implies g(X_{n'_k}) \xrightarrow{\text{a.s.}} g(X)$$

$$g(X_{n'_k}) \xrightarrow{P} g(X)$$

$$g(X_n) \xrightarrow{P} g(X)$$